

Curie Second-Filter Removal: Quick Results Comparison

This compares the GitHub version with the second field-stabilization filter against the current local version without that filter.

Definitions:

- With filter: HEAD:curie/proc.py, using the field-ready cut and a common retained field window.
- Without filter: current local curie/proc.py, using all loops and Method II evaluated per loop at $0.98 H_{\max}(T)$.

Bottom Line

Quantity	With filter	Without filter	Change
Series A weighted method mean	213.9 K	197.7 K	-16.2 K
Report headline uncertainty	+/- 12.3 K	+/- 18.3 K	+6.0 K
Conservative all-drive envelope	+/- 19.6 K	+/- 26.1 K	+6.5 K

Compared with the old report prose value 216(10) K, the updated no-filter report value is 198(18) K. The center moves down by about 18 K, and the uncertainty increases.

Series A Half-Height Methods

Method	With filter	Without filter	Change
M0	216.4 +/- 0.8 K	196.9 +/- 1.8 K	-19.5 K
Msat	214.8 +/- 1.0 K	204.0 +/- 1.5 K	-10.8 K
M0_ext	199.1 +/- 1.7 K	169.4 +/- 3.3 K	-29.7 K

All-Series Method Means

Series	With filter	Without filter	Change
A	213.9 K	197.7 K	-16.2 K
B	202.9 K	196.0 K	-6.9 K
C	236.5 K	229.1 K	-7.3 K

Data Included

Series	With filter loops and T range	Without filter loops and T range	Added loops
A	141 loops, 180.8-273.3 K	172 loops, 82.9-273.3 K	+31
B	155 loops, 159.1-273.2 K	179 loops, 83.1-273.2 K	+24
C	177 loops, 221.3-273.3 K	251 loops, 110.7-273.3 K	+74

Interpretation

Removing the second filter adds the early low-temperature loops back into the normalization. That changes the dynamic range of each normalized magnetization curve, so the half-height crossing shifts substantially lower.

The shift is largest in Series A and especially in the extrapolated-tail method ($M0_{\text{ext}}$). The no-filter version is therefore more conservative but less internally consistent: the method spread in Series A grows from 9.5 K to 18.3 K, which drives the larger final uncertainty.

Short version: the filtered version gives a higher and tighter transition estimate; the no-filter version gives a lower, broader estimate because it includes the early low-temperature loops instead of discarding them as field-settling data.